

Mathematics for Social Scientists

Part 3: Linear Algebra III - Matrices and Linear Maps

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Outline

Matrix calculus and linear maps

Matrix Inverse

Matrix Rank

Linear Independence

Determinant

Linear Functions

Eigenvalues and Eigenvectors

The inverse of a matrix

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix.

Definition 1.

A matrix $B \in \mathbb{R}^{n \times n}$ satisfying

$$AB = BA = I_n$$

is called the **inverse (matrix)** of A . Notation: $B = A^{-1}$.

- The notion of an inverse is only defined for square matrices.
- The inverse of a matrix does not have to exist.
- If it exists, the inverse is uniquely determined. In this case, A is called **invertible** or **non-singular** (opposite: **singular**).

Two questions

Example.

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}. \text{ The inverse is given by } A^{-1} = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}.$$

Two questions

- How can we determine whether a matrix A is invertible?
- How can we compute the inverse?

Let us first consider the second question!

Computation of the inverse matrix

Consider the invertible matrix $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 4 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$.

Form the following **augmented matrix (Gauss tableau)** where the right-hand side is the identity:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 3 & 4 & 1 & 0 & 0 & 1 \end{array} \right).$$

- Reduce the left block to I_3 using elementary row operations **and apply the same operations** to the right block.

Computing A^{-1} step by step (Gauss-Jordan)

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 4 & 1 \end{pmatrix}, \quad \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 3 & 4 & 1 & 0 & 0 & 1 \end{array} \right).$$

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - 3R_1 \implies \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & -2 & 1 & -3 & 0 & 1 \end{array} \right)$$

$$R_3 \leftarrow R_3 - 2R_2 \implies \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right)$$

$$R_2 \leftarrow -R_2 \implies \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right)$$

$$R_1 \leftarrow R_1 - 2R_2 \implies \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 2 & 0 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right)$$

Computation of the inverse matrix ctd.

In our specific example, this yields:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 2 & 0 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right).$$

- Now the inverse is the matrix on the right-hand side of the Gauss tableau.

That is, the inverse of the matrix $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 4 & 1 \end{pmatrix}$ is

$$A^{-1} = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -1 & 0 \\ 1 & -2 & 1 \end{pmatrix}.$$

Using Inverse to solve the SLE

In order to solve the SLE $Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ one can proceed as follows:

$$\begin{aligned} Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} &\Leftrightarrow A^{-1}Ax = A^{-1} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \Leftrightarrow x = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -1 & 0 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\ &\Leftrightarrow x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

Rank of a matrix

Definition 2.

The **rank of a matrix** $A \in \mathbb{R}^{r \times n}$ is the rank of the SLE $Ax = b$ (formerly written as $(A|b)$) for some $b \in \mathbb{R}^r$.

Notation: $\text{rank}(A) = 2$.

Remark.

This does not depend on the choice of b .

Example.

The rank of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 0 \end{pmatrix}$ is equal to 2.

Alternative characterisation via **linear independence!**

Linear independence

Let's imagine you have a collection of vectors. Can you write one of them as a linear combination (addition / scalar multiplication) of others?

If no, then they are *linearly independent*.

If yes, then they are *linearly dependent*.

Example.

- The vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly independent.
- The vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are linearly dependent.
- The vectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$ are linearly dependent.

Linear Independence ctd

Definition 3 (Linear Independence).

A finite family $v_1, \dots, v_r$ of vectors from $\mathbb{R}^n$ is called **linearly independent** if:

$$\lambda_1 v_1 + \dots + \lambda_r v_r = \mathbf{0} \quad \Rightarrow \quad \lambda_1 = \dots = \lambda_r = 0$$

Otherwise the family is called **linearly dependent**.

Remark.

- A single vector $v \in \mathbb{R}^n$ is linearly independent if and only if $v \neq \mathbf{0}$,
- The vector $\mathbf{0}$ is linearly dependent on any other vector.
- Any family containing the same vector twice is linearly dependent.
- If $r \geq 2$, the vectors $v_1, \dots, v_r$ are linearly dependent, if one v_i is the linear combination of the other vectors.

Linear dependence and rank

Proposition 1.

The rank of a matrix $A^{r \times n}$ is the **maximal** number of linear independent row vectors (considered as elements of $\mathbb{R}^n$).

Example.

$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 4 \\ -2 & -3 & 2 \end{pmatrix}$ has maximal two linearly independent row vectors. The first two rows are linearly independent, whereas all three are not. This is because, third row can be written as a linear combination of the first two $((-2) \cdot r_1 + r_2)$. Therefore, it's rank is 2.

Proposition 2.

Let $A \in \mathbb{R}^{r \times n}$. Then

$$\text{rank}(A) = \text{rank}(A^T).$$

Rank and existence of an inverse

Now: **When does the inverse of a matrix A exist?**

Theorem 3.

A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if

$$\text{rank}(A) = n.$$

Corollary 4.

A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if the SLE

$$Ax = b$$

has a unique solution for every $b \in \mathbb{R}^n$.

Easy criterion to determine whether or not a matrix is invertible: *determinant!*

The determinant of a matrix

Notation: for a matrix $A \in \mathbb{R}^{n \times n}$ we write $A = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$,

where $\mathbf{a}_i$ is the i -th column vector.

Definition 4.

A **determinant** (of order n) is a function

$$\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}, \quad A \mapsto \det(A),$$

such that the following properties are satisfied:

i. For all $s, t \in \mathbb{R}$ and all $i = 1, \dots, n$,

$$\begin{aligned} \det(\mathbf{a}_1, \dots, \mathbf{a}_{i-1}, s\mathbf{a}_i + t\mathbf{a}'_i, \mathbf{a}_{i+1}, \dots, \mathbf{a}_n) \\ = s \cdot \det(\mathbf{a}_1, \dots, \mathbf{a}_{i-1}, \mathbf{a}_i, \mathbf{a}_{i+1}, \dots, \mathbf{a}_n) + t \cdot \det(\mathbf{a}_1, \dots, \mathbf{a}_{i-1}, \mathbf{a}'_i, \mathbf{a}_{i+1}, \dots, \mathbf{a}_n) \end{aligned}$$

ii. $\det(A) = 0$ if $\mathbf{a}_i = \mathbf{a}_j$ for some $i \neq j$,

iii. $\det(I_n) = 1$.

Determinants ctd

Theorem 5.

For all $n \geq 1$, there exists exactly one determinant of order n .

One can explicitly define the determinant for arbitrary n .

On the following slides, we will only give formulae for the cases $n = 2$ and $n = 3$.

Theorem 6.

A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if $\det(A) \neq 0$.

The determinant of order 2

Proposition 7.

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The determinant of order two is given by the formula

$$\det(A) = ad - bc.$$

Example.

$$\det \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = -2.$$

A closed formula for the inverse of a 2×2 matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be invertible, that is, $\det(A) = ad - bc \neq 0$. Then, the inverse A^{-1} is given by the formula

$$A^{-1} = \frac{1}{ad - bc} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Exercise: Check whether the following matrices are invertible and if yes, compute the inverse:

- $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$
- $B = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}$
- $C = \begin{pmatrix} 3 & 3 \\ 1 & 1 \end{pmatrix}$

The determinant of order 3

Proposition 8.

Let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ The determinant of order $n = 3$ is given by the formula

$$\det(A) = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} - a_{31}a_{22}a_{13}.$$

Example.

$$\text{Let } A = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}. \det(A) = ?$$

Some rules for the computation of determinants

Let $A, B \in \mathbb{R}^{n \times n}$.

- Let $A = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$ be diagonal. Then $\det(A) = a_{11} \cdot a_{22} \cdot \dots \cdot a_{nn}$,
- $\text{rank}(A) < n \Leftrightarrow \det(A) = 0$.
- $\det(A \cdot B) = \det(A) \cdot \det(B)$.
- In particular, in the case that A is invertible, $\det(A^{-1}) = (\det(A))^{-1}$.
- $\det(A) = \det(A^\top)$.

Linear mappings and matrices

Definition 5 (Linear map / linear transformation).

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if

- $T(x + y) = T(x) + T(y)$ for all $x, y \in \mathbb{R}^n$,
- $T(\lambda x) = \lambda T(x)$ for all $\lambda \in \mathbb{R}, x \in \mathbb{R}^n$.

Matrix representation

For any matrix $A \in \mathbb{R}^{m \times n}$,

$$T_A(x) := Ax \quad (x \in \mathbb{R}^n)$$

is a linear map. Conversely, every linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is of the form T_A for some matrix A (“linear map” = “linear transformation”).

Some connections between the things we have learned up to now

Let $A \in \mathbb{R}^{n \times n}$. Then we have:

A has full rank equal to n

$$\Leftrightarrow \det(A) \neq 0$$

$$\Leftrightarrow A \text{ is invertible}$$

$$\Leftrightarrow Ax = b \text{ has a unique solution for every } b \in \mathbb{R}^n$$

$$\Leftrightarrow f_A \text{ is bijective}$$

$$\Leftrightarrow \text{the columns of } A \text{ are linearly independent}$$

$$\Leftrightarrow \text{the rows of } A \text{ are linearly independent}$$

Eigenvalues and eigenvectors

Definition 6.

Let $A \in \mathbb{R}^{n \times n}$.

- $\lambda \in \mathbb{R}$ is called **eigenvalue** of A if there exists $\mathbf{0} \neq v \in \mathbb{R}^n$ such that

$$A \cdot v = \lambda \cdot v.$$

- Every $v \neq \mathbf{0}$ satisfying $A \cdot v = \lambda \cdot v$ is called **eigenvector** (for the eigenvalue λ).

Eigenvalues and Eigenvectors ctd

Example.

The matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 3$ with the corresponding examples of eigenvectors:

$$x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{and} \quad x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Exercise: Check this!

Solution

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$P_A(\lambda) = \det(A - \lambda I_2) = \det\begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3),$$

so the eigenvalues are $\lambda = 1, 3$.

$$\lambda = 1: (A - I_2)v = 0 \Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} v = 0 \Rightarrow v = t \begin{pmatrix} 1 \\ -1 \end{pmatrix}, t \neq 0.$$

$$\lambda = 3: (A - 3I_2)v = 0 \Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0 \Rightarrow v = t \begin{pmatrix} 1 \\ 1 \end{pmatrix}, t \neq 0.$$

Hence (up to scaling) $x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ for $\lambda_1 = 1$ and $x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda_2 = 3$.

Getting the eigenvalues

Q: How to determine eigenvalues and eigenvectors?

Proposition 9.

For $A \in \mathbb{R}^{n \times n}$ the following statements are equivalent:

- λ is an eigenvalue of A .
- $\det(A - \lambda \cdot I_n) = 0$.

Thus, the search for eigenvalues can be reduced to the question of finding the roots/zeros of the so-called *characteristic polynomial*:

$$P_A(\lambda) := \det(A - \lambda \cdot I_n).$$

Remark.

$P_A(\lambda)$ is a polynomial of degree n .

Examples

- $A = \begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix} \rightsquigarrow P_A(\lambda) = \lambda^2 - 3\lambda + 2.$

- $B = \begin{pmatrix} 0 & -1 & 1 \\ -3 & -2 & 3 \\ -2 & 2 & 3 \end{pmatrix} \rightsquigarrow P_B(\lambda) = -\lambda^3 + \lambda^2 + 13\lambda - 13.$

Getting the eigenvectors

- Note that $A \cdot x = \lambda \cdot x \Leftrightarrow (A - \lambda \cdot I_n)x = \mathbf{0}$.
- Thus, finding an eigenvector for the eigenvalue λ is reduced to the problem of solving the SLE with right-hand side given by the matrix $A - \lambda \cdot I_n$ and left-hand side $b = \mathbf{0}$ (we are only interested in the non-trivial solutions of this SLE).

Getting the eigenvectors ctd

Example (see example from before).

$A = \begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$. For $\lambda_1 = 1$ we get the SLE

$$-2x_1 + 6x_2 = 0$$

$$-x_1 + 3x_2 = 0$$

which has the solution set $\mathbb{L} = \left\{ \begin{pmatrix} 3a \\ a \end{pmatrix} : a \in \mathbb{R} \right\}$.

Thus, for every $a \neq 0$, $\begin{pmatrix} 3a \\ a \end{pmatrix}$ is a eigenvector for the eigenvalue 1. What for $\lambda_2 = 2$?